Md Oli Uz Zaman

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Summary

Packaging R&D Engineer at **Intel** with experience in advanced 2.5D/3D integration and master's research in emerging non-volatile memory systems. Author of five peer-reviewed publications on memristor-based neuromorphic computing. Seeking a PhD to investigate energy-efficient AI architectures that integrate emerging memories and 3D technologies, with an emphasis on reliability.

Research Interests

- Computer Architecture
- Memory Systems
- Emerging Memory Technologies
- AI Hardware Accelerators
- Compute-in-Memory
- Heterogeneous Computing
- RTL and VLSI Design
- Neuromorphic Computing
- Reliability-Aware Computing

Education

- **University of South Alabama** — Alabama, USA
  Master of Science - Electrical Engineering; GPA: 4.00/4.00 — *Jan 2021 - Dec 2022*
- **Rajshahi University of Engineering and Technology (RUET)** — Bangladesh
  Bachelor of Science - Electronics and Telecommunication Engineering; GPA: 3.52/4.00 — *Jan 2012 - Oct 2016*
  Ranked 10th of 60 students

Skills Summary

- **Scripting/Programming**: Python, Perl, Verilog, VHDL, C, MATLAB, SQL, MIPS Assembly
- **EDA** & **Circuit Design Tools**: Cadence Virtuoso, Quartus II, ModelSim, NeuroSim, PSpice
- **Platforms**: Linux, Windows, Arduino, HPC
- **Others**: JMP, SQL Pathfinder, Astah SysML, Overleaf, Waveform, Analog Discovery, Oscilloscope, Signal Generator

Experience

- **Intel** — Oregon, USA
  Packaging R&D Engineer — *Sep 2023 - Present*
  - Support advanced packaging metrology development for 2.5D EMIB and 3D Foveros/Foveros-Direct technologies.
  - Develop inspection recipes for defect detection: warpage, wafer inner crack, misaligned chiplets, epoxy residuals.
  - Partner with process modules and integration to develop DOE to identify root cause of product excursions.
  - Collaborated with high-volume manufacturing teams to maintain and improve metrology production workflows.
  - Engaged in product transfer (such as 18A Foveros) from TD (Technology Development) to other Intel factories.

- **Intel** — California, USA
  NAND Product Quality and Reliability Engineer (PQRE) — *Jan 2023 - Sep 2023*
  - Designed and developed Python scripts to automate large-scale test data processing and generate customized outputs for engineering analysis
  - Owned the ELFR (Early Life Failure Rate) qualification process to assess DPM (Defects Per Million) performance of 3D NAND under customer-like conditions and drive corrective actions
  - Developed PERL scripts and performed Electrical Failure Analysis (EFA) on bench setups to isolate faults, segment defect modes, and provide feedback to the Fab for process improvement pathfinding.

- **University of South Alabama** — Alabama, USA
  Graduate Assistant — *Jan 2021 - Dec 2022*
  - **Research Assistant (RA) Responsibility**:
    - Research on **Memristor/RRAM Based Neuromorphic Computing**
    - Utilized a cutting-edge 40 nm *Ag:a-Si* memristor device for real-world behavior extrapolation
    - Restored 90% inference accuracy from a Stuck-At-Fault (SAF) defected memristor crossbar with 10% initial accuracy
    - Eliminated energy, latency, and hardware overhead while enhancing immunity to SAF defects and non-linearity
  - **Teaching Assistant (TA) Responsibility**:
    - Graded lab projects and reports; Assisted undergraduate students with laboratory experiments
    - Teaching Assistant Courses: Digital Logic Design Lab, Microprocessor Sys Interface Lab, Programmable Logic Devices Lab, Embedded System Design Lab, Electronics Lab

- **ZH Sikder University of Science and Technology** — Bangladesh
  Lecturer, Electrical Engineering — *October 2019 - Dec 2020*

  TMSS Engineering College — Bangladesh
  Lecturer, Electrical Engineering — *May 2018 - October 2019*
  - **Courses Taught**: VLSI Design, Digital Electronics, Electrical Circuit Analysis, Electronic Devices and Circuit Theory, Control Systems, Electromagnetic Fields and Waves, Radio and TV Engineering, Telecommunication Engineering, Satellite and Radar Communication, Wireless Communication

Relevant Graduate Coursework

- **MS**: HDL Logic Synthesis, VLSI, Digital Computer Architecture, Advanced Computer Architecture, Embedded Systems

Projects

- **Recipe Change Tracker – Python Desktop Software (Intel Corporation - Metrology Module)**: Built a lightweight GUI-based desktop application to track recipe changes across the Metrology Module. Enabled real-time multi-user access to a centralized CSV log over a networked server. Key features include calendar date picker, input validation, recipe-based filtering, scrollable history view, and standalone .exe packaging using PyInstaller
- **8-Bit ALU Design**: Achieved 10% latency improvement in an 8-bit ALU using VHDL for arithmetic and logic operations; conducted synthesis, timing, and functional simulations
- **128-Byte Memory Design**: Designed synchronous memory using behavioral-level VHDL with read/write operations on rising clock edges; evaluated synthesis and performed comprehensive timing and functional simulations
- **Cadence Virtuoso Project**: Schematic design and analysis of a 4×4 SRAM memory architecture
- **Brain Tumor Detection & Analysis using Morphological Operations in Matlab**: Implemented image segmentation and morphological operations to detect brain tumors and estimate size and growth rate
- **Practicum – Smart Coffee Maker**: Astha SysML was employed for designing and analyzing the mind map, concept of operation, stakeholder requirements, stakeholder desirements, stakeholder diagram, use case, block definition diagram, and requirement diagram of a smart coffee maker

Publications

- **Journal**: **M Oli-Uz-Zaman**, SA Khan, W Oswald, Z Liao, J Wang, "Stuck-At-Fault Immunity Enhancement of Memristor-Based Edge AI Systems," IEEE Journal on Emerging and Selected Topics in Circuits and Systems, Sept 2022.
- **Journal**: **M Oli-Uz-Zaman**, SA Khan, G Yuan, Z Liao, J Fu, C Ding, Y Wang, J Wang, "Mapping Transformation Enabled High-Performance and Low-Energy Memristor-Based DNNs," Journal of Low Power Electronics & Applications, Mar 2022.
- **Journal**: SA Khan, **M Oli-Uz-Zaman**, J Wang, "PAWN: Programmed Analog Weights for Non-linearity Optimization in Memristor-based Neuromorphic Computing System," IEEE Journal on Emerging and Selected Topics in Circuits and Systems, January 2023.
- **Conference**: **M Oli-Uz-Zaman**, SA Khan, G Yuan, Y Wang, Z Liao, J Fu, C Ding, J Wang, "Reliability Improvement in RRAM-based DNN for Edge Computing," IEEE International Symposium on Circuits and Systems (ISCAS), May 2022.
- **Conference**: **M Oli-Uz-Zaman**, SA Khan, W Oswald, Z Liao, J Wang "Reconfigurable Mapping Algorithm based Stuck-At-Fault Mitigation in Neuromorphic Computing Systems," Great Lakes Symposium on VLSI, June 2023.
- **Conference**: H Sarker, **M Oli-Uz-Zaman**, IH Chowdhury, SAH Chowdhury, MR Islam, "Node Estimation Approach of Underwater Communication Networks Using Cross-correlation for Direct and Multi-path Propagation," IEEE International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), January 2019.

Selected Research Presentations

- Poster Presentations on **Advanced Defect Detection Methodology for EMIB-T Bridge Structures**, Intel Foundry Spark Sessions, 2026
- IEEE International Symposium on Circuits and Systems, Austin, TX
- 31st ACM SIGDA University Demonstration, NDSU
- 4th IBM IEEE CAS/EDS AI Compute Symposium (AICS'21)
- University of South Alabama Annual Graduate Research Forum
- IEEE Technical Meeting, University of South Alabama Annual

Invited Talks & Lectures

- **Invited Lecturer, Advanced VLSI (two lectures delivered)**, University of Alabama, Tuscaloosa – Fall 2026
- **E-USem: Experiential-learning-based Undergraduate Semiconductor Workforce Exploration**, University of South Alabama – Fall 2024
- **Higher Studies and Career Directions in the USA**, RUET, Bangladesh – October 2025

Reviewer Service

- Frontiers in Neuroscience – 2026
- IEEE International Midwest Symposium on Circuits and Systems (MWSCAS) – 2023, 2026.
- Great Lakes Symposium on VLSI (GLSVLSI) – 2026
- IEEE SoutheastCon – 2022

Honors and Awards

- OATD Department Achievement Award, Intel – Three-time recipient (2025–2026)
- Intel OATD Metrology Award (2023–2026)
- Recognized for outstanding performance (QPB+) in 7 of the last 12 quarters
- Warren & Kathy Nicholson Endowed Scholarship, University of South Alabama – Fall 2022
- Tau Beta Pi – Fall 2022
- Phi Kappa Phi – Fall 2022
- Conference Travel Fellowship for IEEE ISCAS – Summer 2022
- Technical Scholarship for Outstanding Academic Result at RUET (2012–2016)